



Polypropylene mesh coated with hyaluronic acid/polyvinyl alcohol composite hydrogel for preventing bowel adhesion

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ABSTRACT

Polypropylene (PP) mesh is the most widely used prosthetic material in hernia repair. However, the efficacy of implanted PP mesh is often compromised by adhesion between viscera and PP mesh. Thus, there is a recognized need for developing an anti-adhesive PP mesh. Here, a composite hydrogel coated PP mesh with the prevention of adhesion after hernia repair was designed. The composite hydrogel coating was prepared from polyvinyl alcohol (PVA) and hyaluronic acid (HA) by using the freezing-thawing (FT) method. To overcome the shortcoming of the long time of the traditional freezing-thawing method, a small molecule 3,4-dihydroxyphenylacetic acid (DHPA) was introduced to promote the formation of composite hydrogel. The as-prepared composite hydrogel coating displayed modulus more closely resembling that of native abdominal wall tissue. *In vitro* studies illustrated that the resulting meshes showed excellent coating stability, hemocompatibility, and non-cytotoxicity. *In vivo* experiments using a rat abdominal wall defect model demonstrated that the composite hydrogel coated PP mesh could prevent the formation of adhesion, alleviate the inflammatory response, and reduce the deposition of collagen around the damaged tissue. These disclosed results manifested that the PP mesh coated with HA/PVA composite hydrogel might be a promising application in preventing adhesion for hernia repair.

1. Introduction

Hernia surgery is one of the most common diseases in the world with about 20 million repairs are operated each year [1]. Currently, tension-free hernia repair and implantation of prosthetic materials have become the most commonly used treatments [2]. Polypropylene (PP) mesh with the property of good stability, inherent inert chemical, excellent non-cytotoxicity and so on, which has been employed as the most popular prosthetic materials in hernia repair. However, the application of PP mesh may induce severe abdominal adhesion because of its weight, hydrophobic surface, and foreign body feeling. In severe cases, abdominal adhesions could lead to a series of complications, including intestinal obstruction, bowel erosion, infertility, intestinal fistula, and even cumbersome reoperation, seriously affecting the quality of life of patients and increasing the financial burden [3,4]. Therefore, it is urgent to enhance anti-adhesive property of PP mesh.

In recent years, numerous approaches have been applied to prevent the formation of adhesion, such as creating physical barriers, chemical barriers, biological barriers, as well as pharmaceutical treatment [5–7]. The construction of physical barriers integrated with PP mesh is the simplest and the most direct method to inhibit adhesion. An ideal physical barrier can not only completely isolate damaged tissue and the surrounding tissue but also match the mechanical properties of the natural abdominal wall tissue. Hydrogel, a family of polymeric materials with crosslinked hydrophilic networks and high water content, has gained much attention in resisting postoperative adhesion [8,9]. In a liquid environment, the water molecules adsorbed on the hydrophilic hydrogel surface create a hydration layer that prevents protein adsorption. Cells cannot directly adhere to the hydrogel surface because its adhesion depends on protein adsorption [10,11]. Thereby, the formation of adhesion is inhibited.

Synthetic or natural polymers or a combination of them (hybrid

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synthetic and natural polymers) are used to prepare most hydrogel barriers. Polyvinyl alcohol (PVA), a semi-crystalline hydrophilic polymer, exhibits the unique comprehensive properties such as water solubility, biocompatibility, and nontoxicity. Owing to these excellent properties, PVA has been widely applied in biomedical fields [12–14]. PVA hydrogels with physical crosslink network can be obtained through the freezing-thawing (FT) cycle method. In this process, the generated hydrogen bonds and crystalline clusters play a crucial role in establishing the physical crosslink network structure of PVA hydrogels [15–17]. However, the traditional FT cycle method is highly dependent on lower freezing temperature, a long cycle time, and high numbers of cycles. Therefore, new approaches need to be developed to prepare PVA hydrogel under relatively simple and mild conditions. A recent study by Kaneko et al. [18] has proposed that the addition of small molecules with carboxyl groups and phenolic hydroxyl groups can form hydrogen bonds between PVA chains and accelerate the formation of crystal regions in the PVA hydrogels under mild conditions.

Hyaluronic acid (HA) is a natural and water-soluble anionic polysaccharide consisted of D-glucuronic acid and N-acetyl glucosamine, which is one of the most crucial and common glycosaminoglycans in vertebrate bodies. Given the inherent moisture retention, biocompatibility, and biodegradability, HA-based hydrogels have been applied in many fields, such as post-operative anti-adhesion [19], wound healing [20], drug delivery [21], and osteoarthritis treatment [22]. However, HA-based hydrogels can degrade in the human body [23] and have relatively weak mechanical properties [24], which limit their potential application. To fabricate HA-based hydrogels with suitable mechanical properties, the use of chemical cross-linking agent or organic solvent is unavoidable. This may impair the biocompatibility of HA-based hydrogels, especially in biomedical applications. Notably, one study by Zhang et al. [25] suggested the use of the FT cycle method to induce hyaluronic acid gelation. This method can ensure excellent

biocompatibility of HA-based hydrogels but is not effective in terms of time and cost because it requires lower freezing temperature, a long cycle time, and high numbers of cycles.

An ideal adhesion preventive hydrogel barrier should remain at the damaged site during the critical period of adhesion formation, causing minimal inflammatory response or foreign body reaction [26–28]. In view of the above, the PVA and HA were deposited as a composite hydrogel barrier on the surface of the PP mesh by the FT cycle method. At the same time, 3,4-dihydroxyphenylacetic acid (DHPA), a biologically active small molecule [29–31], with carboxyl groups and phenolic hydroxyl groups is introduced to induce the formation of hydrogen bonds between the macromolecular chains and reduce the freezing time. Further, the mechanical property and stability of the HA/PVA composite hydrogel can be regulated by varying the amount of DHPA.

Therefore, this study presented a simple and benign FT method to develop an anti-adhesive PP mesh with a hydrophilic HA/PVA composite hydrogel coating (Fig. 1). The structure of hydrogel coating was performed by field emission scanning electron microscope (FE-SEM), X-ray diffraction (XRD), and thermogravimetric analysis (TG). The mechanical properties, swelling behavior and biodegradation *in vitro* of the hydrogel coating were also investigated in detail. Besides, surface morphology, surface wettability, and coating stability of the modified PP mesh were evaluated. Further, the mouse fibroblast L929 cell line was selected as the tested cell line to assess the cytocompatibility of the hydrogel coating and PP-based meshes. Finally, the anti-adhesive property *in vivo* of the modified PP mesh was studied by using a rat abdominal wall defect model. This study may provide a novel and efficient method for the preparation of anti-adhesive PP hernia mesh.

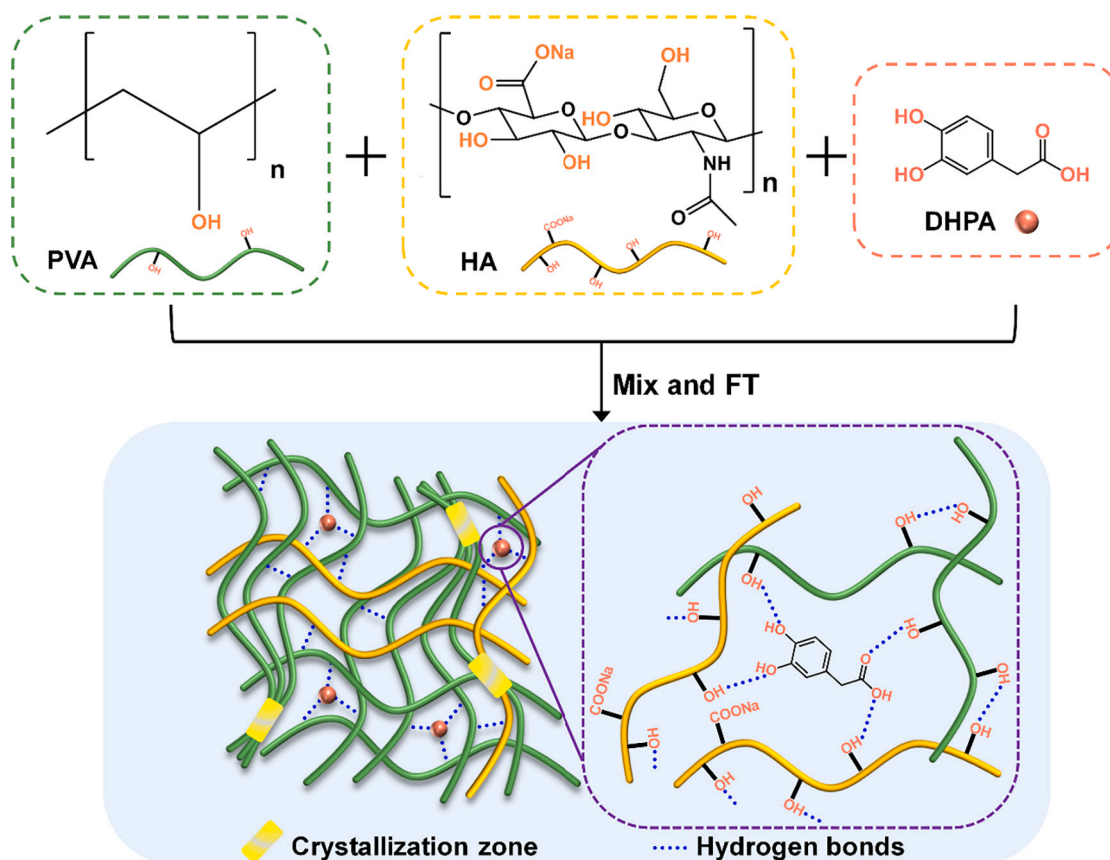


Fig. 1. The schematic diagram of preparation of the anti-adhesive hydrogel coating.

2. Materials and methods

2.1. Materials

Polyvinyl alcohol (PVA) (polymerization degree of 1750 ± 50), ethanol, and dichloromethane were purchased from China National Pharmaceutical Group Co., Ltd. (China). Hyaluronic acid sodium (HA), hyaluronidase, and 3,4-dihydroxyphenylacetic acid (DHPA) were purchased from Aladdin Industrial Inc. (Shanghai, China). The polypropylene (PP) meshes were kindly provided by Donghua University. Mouse fibroblast L929 was obtained from the Stem Cell Bank of Chinese Academy of Sciences (China). Phosphate buffered saline (PBS, pH 7.4), Dulbecco's modified eagle medium (DMEM), 0.25% trypsin-EDTA, CCK-8 cell viability assay kit and live/dead fluorescent staining cell kit (FDA/PI) were purchased from KeyGen Biotechnology Co., Ltd. (Jiangsu, China). Fetal bovine serum (FBS) was obtained from Gibco Company (USA). All other chemical agents were commercially available and used without any further processing. The experimental water was deionized water.

All the animals were supplied by the Laboratory Animal Centre of Southeast University (Nanjing, China). The procedures involving animals and their care were performed in accordance with the NIH Guidelines (NIH Pub. No. 85-23, revised 1995). All animal experiment protocols were approved by the Institutional Animal Care and Use Committee of Southeast University and were in compliance with all the regulatory guidelines. The assigned approval number is 20230227091 (Approved on February 27, 2023).

2.2. Preparation of modified PP meshes and composite hydrogel

To remove antioxidants and other soluble additives, the PP meshes were washed with a mixture of ethanol (60 mL) and dichloromethane (30 mL) for three times at room temperature. Then, the PP meshes were dried under a vacuum at 30°C for further use. To coat the composite hydrogel coating on the surface of the PP mesh, the precursor solution of hydrogel should not be thick. Hence, HA (0.15 g) was first dissolved in 30 mL deionized water and stirred at 60°C to obtain a 0.5% HA solution. Then, 1.5 g of PVA was added to the HA solution and heated to 80°C with stirring until completely dissolved to obtain the HA/PVA homogeneous solution. Different contents (0%–1.35%) of DHPA was added to the HA/PVA solution under stirring. Subsequently, the mixture solution of PVA, HA, and DHPA was brushed on the surface of PP meshes, and then these PP meshes were frozen at -20°C for 3 h and thawed at room temperature for 4 h to obtain the modified PP meshes. The modified PP meshes were denoted as PVH-c-PP, and PVHD_x-c-PP, where the subscript is the added content of DHPA (Table S1). The virginal PP was denoted as V-PP.

For the preparation of composite hydrogels, the mixture solution of PVA, HA, and DHPA was poured into mold. After freezing at -20°C for 3 h and thawing at room temperature for 4 h, the hydrogel composed of PVA, HA and DHPA was obtained, which were named as PVHD_x, where the subscript x is the content of DHPA (Table S1). Composite hydrogel without the addition of DHPA was named PVH, as a control sample.

2.3. Characterization of surface property

The microstructure of lyophilized composite hydrogels and the surface morphology of PP-based meshes were observed with a field emission scanning electron microscope (FE-SEM, Ultra Plus, Carl Zeiss, Germany). The pore size of lyophilized hydrogels was calculated using Image J. To verify the hydrophilicity changes of all PP meshes, the static water contact angle (WCA) was determined by using a contact angle goniometer (Dataphysics, Germany).

2.4. Mechanical test

A compression test was conducted to test the compressive mechanical properties of the composite hydrogel and natural abdominal wall tissue by using a mechanics analyzer (FLR-303, Flora Automatic Technology, China). The cylindrical hydrogel samples (diameter of 22 mm, height of 14 mm) were prepared to perform this test at the speed of 3 mm/min. The average of at least five tests was reported.

2.5. Swelling and degradation behavior test

The lyophilized composite hydrogel samples were immersed in PBS (pH = 7.4) buffer solution at 37°C until swelling equilibrium. At specific time points, the samples were removed from the solutions and the excess of liquid on the surface was absorbed quickly by filter paper. The swelling ratio (SR, %) was defined as follows ($N = 5$):

$$SR (\%) = \frac{W_t - W_0}{W_0} \times 100 \quad (1)$$

where W_0 is the initial weight of lyophilized hydrogel samples before absorbing in PBS, g; W_t is the weight of swollen hydrogels at the specific time t , g.

For evaluating the degradation kinetics of the composite hydrogel, the samples were immersed in 5 mL PBS (pH = 7.4) containing 50 U/mL hyaluronidase at 37°C [24]. At selected time intervals, the samples were removed from the solution and weighed. The remaining weight ratio was calculated via the following equation:

$$\text{Remaining weight } (\%) = \frac{W_t}{W_0} \times 100 \quad (2)$$

where W_0 is the initial weight of the sample before immersion, g; W_t represents the remaining weight of the sample at time t , g.

2.6. Thermal properties

Thermal property of the composite hydrogel was performed using a Synchronous thermogravimetric analyzer (STA449 F3, NETZSCH) within the temperature range of room temperature to 700°C at a heating rate of $10^\circ\text{C}/\text{min}$ under nitrogen gas. The melting behavior of the composite hydrogel was measured simultaneously also using this analyzer under heating rate of $10^\circ\text{C}/\text{min}$ and nitrogen gas.

2.7. XRD

The lyophilized composite hydrogels were performed on an X-ray diffraction (XRD, Smartlab) with Cu K α radiation in the 2θ range from 5° to 60° with a scan rate of $5^\circ/\text{min}$.

2.8. Coating stability test

The hydrogel coating to prevent the occurrence of postoperative adhesion needs to be stably fixed to the PP mesh surface. The coating stability was reflected through both microscopic morphology and mass change. The modified PP-based meshes were immersed into PBS medium for one month at 37°C , and 40°C . The mass change ratio and residual ratio were used to reflect respectively the mass change of modified PP mesh and the residual mass of hydrogel coating on PP surface after soaking for 30 days.

$$\text{Mass change ratio } (\%) = \frac{W_t - W_0}{W_0} \times 100 \quad (3)$$

$$\text{Residual ratio } (\%) = \frac{W_t}{W_i} \times 100 \quad (4)$$

where W_t is the weight of modified PP meshes at the immersion time t , g;

W_0 is the initial weight of modified PP meshes samples before soaking PBS, g; W_r is the residual weight of hydrogel coating on the PP surface after soaking for 30 days, g; W_i is the initial weight of hydrogel coating on the PP surface, g. The average of at least three tests was reported.

2.9. In vitro biocompatibility evaluation

In *vitro* biocompatibility tests included hemocompatibility test and cytocompatibility test. The hemocompatibility of the different hydrogels and PP-based meshes were evaluated by direct contact methods in vitro. First, fresh rat's whole blood with sodium citrate solution (3.8 wt%) was centrifuged at 1200 rpm for 10 min to separate red blood cells (RBCs). The composite hydrogel samples (25 mg of dry weight) and modified PP-based meshes (size 1.0 cm × 1.0 cm) were balanced in PBS at 37 °C for 12 h. Then, each sample was placed in a centrifuge tube with 5 mL of diluted RBCs suspension (25 µL of RBCs with 4.975 mL of normal saline). The samples were removed from the tubes and centrifuged at 3000 rpm for 5 min after incubation at 37 °C for 1 h. The absorbance of the supernatant was measured at 562 nm with a microplate reader. Deionized water (DIW) and normal saline were used as positive control and negative control. The hemolysis ratio was calculated using the following equation:

$$\text{Hemolysis ratio (\%)} = \frac{OD_s - OD_{nc}}{OD_{pc} - OD_{nc}} \times 100 \quad (5)$$

where OD_s , OD_{nc} and OD_{pc} represent the absorbance of sample groups, negative control groups and positive control groups, respectively. All experiments were conducted in triplicate.

Mouse fibroblast L929 cells were selected for the cytotoxicity assessment of composite hydrogels and PP-based meshes by the CCK-8 method. L929 were cultured in DMEM with 10% FBS and 100 U·mL⁻¹ penicillin/streptomycin at 37 °C in 5% CO₂ for subsequent cell experiments. The hydrogel extraction medium was obtained by immersing the swollen hydrogel in DMEM complete medium at 37 °C for 24 h (200 mg/mL), then the extraction medium was filtered through a 220 µm filter membrane before test. Specifically, cells were collected and resuspended at a density of 2 × 10⁴ cells per mL, and then 100 µL cell suspension per well were seeded in a 96-well plate containing 100 µL hydrogel extraction medium per well. For modified PP, the 200 µL cell suspension were seeded into a 96-well plate containing with sterilized PP mesh sample. The same amount of cell suspension and medium was used as a control. The plates were then placed in an incubator at 37 °C in 5% CO₂. After 1, 3, 5, and 7 days of incubation, CCK-8 assay kit and an FDA/PI fluorescent staining kit were carried out to evaluate the cell viability and live/dead status. The fluorescent images were photographed by an inverted microscope (MShot, MF52, China), and the result of cell viability was recorded by a Microplate Reader (Multiskan FC, Thermo, America) at 450 nm. The cell viability was calculated according to the following equation:

$$\text{Cell viability (\%)} = \frac{A_s - A_b}{A_c - A_b} \times 100 \quad (6)$$

where A_s is the absorbance of sample groups, A_b is the absorbance of blank groups, A_c is the absorbance of control groups. The average of at least three tests was reported.

2.10. Animal test

Thirty Sprague-Dawley rats were randomly divided into six groups in terms of Table S2 and stayed on standard conditions with 25 °C, relative humidity 50–60%, 12 h light/12 h dark. All animals were treated humanely during the experiment.

Under the aseptic condition, the SD rats were anesthetized first and then the abdomen area was shaved, and the skin was disinfected with 0.5% povidone-iodine. Later, a 3 cm midline incision was made, and the

full-thickness defects (1.0 cm × 1.0 cm) of the abdominal wall were created. The defects were then repaired by fixing sterilized mesh samples (native PP mesh and modified PP mesh, size 1.5 cm × 1.5 cm) with 3–0 absorbable surgical sutures. The midline incision was closed with 2–0 absorbable surgical sutures. The animals were sacrificed after 7, 14, and 30 days of implantation, and the abdomen was reopened to assess adhesions according to an adhesion scoring system (Table S3). Subsequently, the full thickness of the abdominal wall of the rat defect was removed and stained by Hematoxylin and eosin (H&E) and Masson trichrome (MT) for further histological evaluation. The anti-TNF-α and anti-IL-6 were used to investigate the inflammatory response of the mesh site. Image J software was used for semiquantitative analysis of TNF-α and IL-6.

2.11. Statistical analysis

All data were expressed as mean ± standard deviation (SD) and were analyzed using IBM SPSS 23.0 software. Significant differences between the groups were made using one-way ANOVA followed by HSD test. $P < 0.05$ (*) between two groups was considered as significant level.

3. Results and discussion

3.1. Characterization of composite hydrogel

The FT method is the most widely used to obtain PVA hydrogel, in which a physical crosslinked structure could be formed through inter- or intramolecular hydrogen bonds in the presence of crystalline regions [32,33]. However, it takes a long time to prepare the hydrogel by FT method at −20 °C (ca. 8–20 h) [34–36]. When the freezing time was reduced to 3 h, PVA, HA, PVA with DHPA, and HA with DHPA could not form hydrogel after freezing at −20 °C and thawing at room temperature (Fig. S1a). Surprisingly, the PVH and PVH with DHPA can form hydrogel (Fig. S1b). The microstructure of composite hydrogels was investigated by SEM. As shown in Fig. 2a, all hydrogels possessed continuous and three-dimensional network structure and the average pore size of these micro-pores was in a range of 41–51 µm. This type of structure not only facilitated the energy dissipation of hydrogels during deformation, but also provided a good platform for cell migration and transportation of nutrients [37].

Mechanical property is a key parameter for judging the performance of hydrogel for its biomedical application. Compression tests were carried out and the results were shown in Figs. 2c, d, and S2. As compared to PVH, the compressive strength of composite hydrogels was not enhanced when the content of DHPA was 0.17%, 0.34%, and 0.51% (Fig. S2a and b). When the content of DHPA was further increased, the composite hydrogels showed higher compressive strength compared with the PVH hydrogel. In Fig. 2c, the compressive strength of PVH hydrogel was 69.7 kPa., while the compressive strength of PVHD₇ hydrogel reached 226.8 kPa, an increase of 225.4% of this value of the PVH hydrogel. This may be attributed to the fact that the hydroxyl and carboxyl groups in DHPA acting as cross-linking points induced hydrogen bonds among the PVA and HA molecular chains [38]. Phenolic hydroxyl groups are more likely to form hydrogen bonds than saturated hydroxyl compounds. The bond energy of hydrogen bonds formed by phenolic hydroxyl groups is higher than that of hydrogen bonds formed by saturated hydroxyl compounds [39]. These hydrogen bonds act as physical cross-linking points, resulting in a stronger network structure. Through the result of FTIR, the generated hydrogen bonds were proved, as shown in Fig. S1c. For PVH hydrogel, the absorption peak of —OH was at around 3471 cm⁻¹. With increased of DHPA content, this peak showed red shift from 3471 cm⁻¹ to 3453 cm⁻¹. The red shift of the —OH groups confirmed the enhancement of the hydrogen bonds. However, with the content of DHPA increasing to 1.35%, the compressive strength of the hydrogel tended to decrease. Excessive hydrogen bonds in PVHD₈ hydrogel led to an increase in the number of cross-

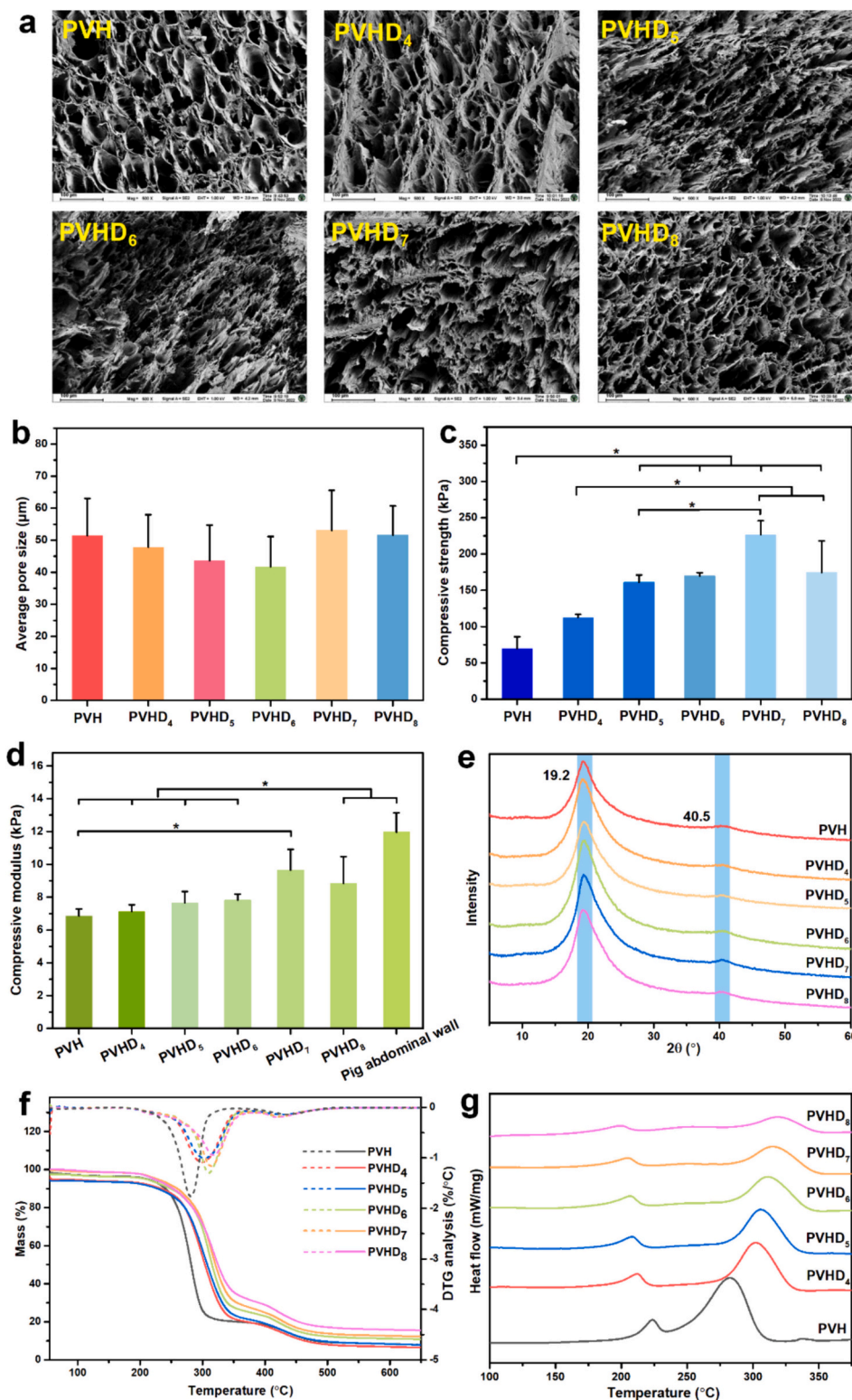


Fig. 2. (a) SEM images and (b) average pore size of lyophilized different composite hydrogels. (c) Compressive strength of different composite hydrogels. Asterisks show the results that are statically significant with P -values < 0.05 (*). Scale bars represent 100 μm . (d) Compressive modulus of different composite hydrogels and the abdominal wall of pig. Asterisks show the results that are statically significant with P -values < 0.05 (*). (e) X-ray patterns, (f) TG-DTG curves (the solid lines are TG and the dashed lines are DTG), and (g) differential scanning calorimetry (DSC) curves of different composite hydrogels.

linking points, which reduced the mobility of the crosslinked grid and prevented the load from being efficiently transferred, resulting in a decrease in compressive strength. It is worthwhile mentioning that the compressive strength of PVHD₈ hydrogel still can reach 174.6 kPa,

which is 150.5% higher than that of the PVH hydrogel.

Further, the freezing time was increased to 6 h, 9 h, and 12 h to prepare PVH and PVHD₇ hydrogel. When the freezing time was 9 h, the compressive strength of PVH and PVHD₇ hydrogel reached the highest

values of 141.2 kPa and 270.4 kPa, respectively (Fig. S2c). It's worth noting that the compressive strength of PVHD₇ hydrogel was constantly higher than that of PVH, no matter how long they were frozen. Moreover, the compressive strength of PVH hydrogel obtained by freezing for 9 h was significantly lower than the compressive strength of PVHD₇ hydrogel obtained by freezing for 3 h. This again confirmed that the addition of DHPA can induce the formation of hydrogen bonds, and these hydrogen bonds can form robust network structure. A similar trend was observed for the compressive modulus of the composite hydrogel. In addition, the compressive modulus of pig abdominal wall was also evaluated. As shown in Fig. 2d, there was no significant difference in compressive modulus between the PVHD₇ hydrogel and the pig abdominal wall ($P > 0.05$). As expected, these results indicated that the composite hydrogel coating exhibited similar hardness to the natural abdominal wall.

XRD characterization was performed to analyze the crystal structure of composite hydrogels. As illustrated in Fig. 2e, all composite hydrogels showed typical crystalline peaks at $2\theta = 19.2^\circ$ and 40.5° , assigning to the lattice planes of (10 $\bar{1}$) and (102) [40]. To evaluate the thermal stability of the composite hydrogels, these hydrogels were performed to thermogravimetric analysis test under nitrogen gas atmosphere. In Fig. 2f, it can be seen that the PVHD₄₋₈ hydrogels exhibited higher residual mass percent than PVH hydrogel. The DTG curves showed that the maximum thermal decomposition rate of PVH hydrogel appeared at 281 °C. Compared to PVH hydrogel, the maximum thermal decomposition rate occurred at 298 °C, 302 °C, 310 °C, 313 °C, and 318 °C for PVHD₄, PVHD₅, PVHD₆, PVHD₇, and PVHD₈ respectively, which is higher than that of PVH hydrogel. These results suggested that the hydrogen bonds between the molecular chains can enhance the thermal stability of the composite hydrogel. In addition, the DSC test was carried out for better knowledge of the thermal stability of the composite hydrogels (Fig. 2g). The distinct endothermic peaks were observed within the 280–320 °C range for all the hydrogels. Furthermore, with increasing DHPA content, the endothermic peak became narrower and shifted to high temperature. This clearly demonstrated the stronger intermolecular forces with higher content DHPA, validating the FTIR results.

3.2. Swelling and degradation behavior of composite hydrogel

Stability in a solution of the composite hydrogels was investigated to determine their potential application as an anti-adhesive coating. Swelling kinetics curves of lyophilized composite hydrogels in PBS buffer solution (pH 7.4) and the microstructure of the composite hydrogels after swelling equilibrium were shown in Fig. 3a and b, respectively. All of the hydrogels exhibited rapid increase in swelling ratio within the first hour after contact with PBS. After 8 h, the swelling trend slowed down and reached a saturation at 96 h. The equilibrium swelling ratios of hydrogels were related to the addition of DHPA with an increase in this resulting in a decrease in the equilibrium swelling ratio. This may be due to the fact that the formation of hydrogen bonds made a denser network structure, therefore, the expansion of the network was restricted and the swelling ratio decreased. Besides, all the composite hydrogels can keep its integrated shape well and display loose connectivity microstructure.

The degradation behavior of the composite hydrogels was studied in terms of observing the mass change and microscopic morphology. The corresponding results were illustrated in Fig. 3c and d. Compared with the PVH hydrogel, the PVHD₄₋₈ hydrogels had a higher remaining weight after the same degradation days. This was attributed to the formation of hydrogen bonds between the molecular chains with the addition of DHPA, which enhanced the stability of the hydrogel network and limited the penetration of the enzyme degradation solution into the hydrogel network. However, it was worth noting that the remaining weight of PVHD₈ hydrogel was lower than that of PVHD₇ hydrogel. This was related to the microscopic morphology of the hydrogels. The pore walls of the internal pore structure of PVHD₈ hydrogel were thinner than those of PVHD₇ hydrogel, making PVHD₈ more susceptible to erosion by the enzyme degradation solution (Fig. 2a). After 30 days of degradation, the microstructure of these hydrogels was observed by SEM (Fig. 3d). The part of the pore walls of the pores structure in the PVHD₈ hydrogel disappeared, confirming that PVHD₈ hydrogel was more easily degraded than PVHD₇ hydrogel.

3.3. Characterization of PP-based meshes

The different composite hydrogels were employed as coating for

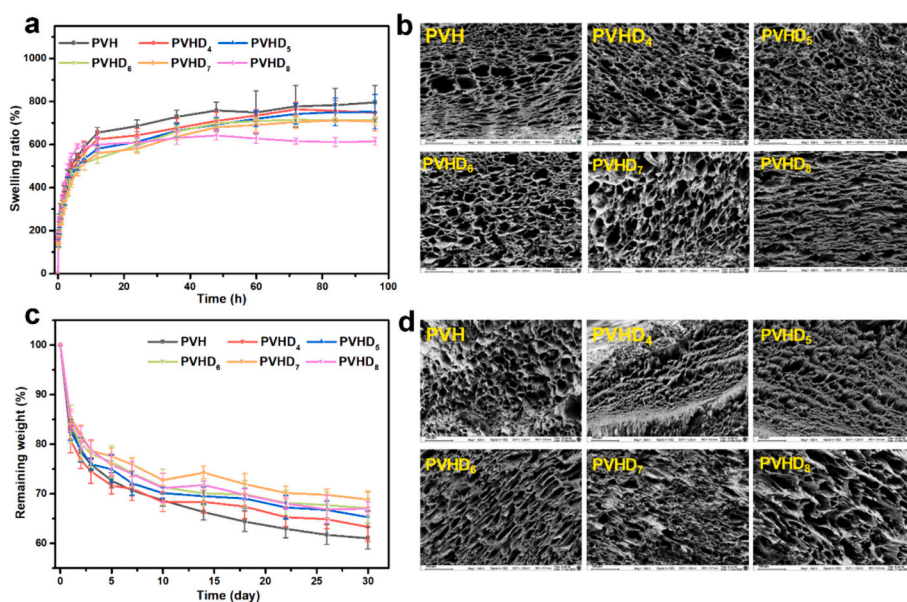


Fig. 3. (a) Swelling behavior of the lyophilized different composite hydrogels in PBS (pH = 7.4) at 37 °C. (b) SEM images of the lyophilized different composite hydrogels after swelling in PBS for 96 h. Scale bars represent 100 μ m. (c) Degradation behavior of the different composite hydrogels in hyaluronidase solution at 37 °C. (d) SEM images of the lyophilized different composite hydrogels after degrading in hyaluronidase solution for 30 days. Scale bars represent 100 μ m.

combination with PP meshes. As shown in the microscopic images (Fig. 4a), these modified PP-based meshes exhibited visible difference in surface and section compared with V-PP mesh. All PP fibers were firmly covered by the composite hydrogel and showed clean and smooth surfaces, especially PVHD₇-c-PP. Moreover, the composite hydrogel coating can also be observed clearly from the SEM images of the cross sections, indicating that composite hydrogel coating can be successfully combined with PP mesh by the FT method. After coated by composite hydrogel, the hydrophilicity of modified PP meshes was significantly improved compared to V-PP (Fig. 4b). The water contact angle (WCA) of V-PP was 112.3°, whereas this value was decreased to 46.7° for PVHD₇-c-PP due to the presence of polar groups in the PVA and HA molecular chains. By controlling the properties of hydrogel coating to match the requirements of the anti-adhesive material, the PVHD₇-c-PP was chosen for subsequent characterization.

In order to effectively reduce adhesion formation, the hydrogel coating should cover the surface of the PP mesh stably to provide sufficient separation between PP mesh and abdominal viscera during or after implantation [41]. Fig. 4c showed the mass change ratio of PVHD₇-c-PP after soaking the PBS medium to simulate the physiological conditions in living organisms. In the initial 1 day, the mass change ratio of PVHD₇-c-PP increased rapidly as the swelling behavior of the hydrogel coating. Subsequently, the change trend was relatively flat as time prolonged. After 30 days, the PVHD₇-c-PP was lyophilized and the weight of the remaining hydrogel coating was determined, while at the same time, the microscopic morphology of lyophilized PVHD₇-c-PP was also observed by SEM. The residual ratio of hydrogel coating mass was

53.9% and 58.6% after 30 days immersing in PBS at 37 °C and 40 °C. Hyaluronic acid (HA) is a natural linear polymer and readily degrades in the human body through hydrolytic or enzymatic reactions [23]. Fortunately, the hydrogen bonds between the PVA and HA molecular chains improve the stability of the hydrogel coating. Besides, it was apparent from Fig. 4e and f that the remaining hydrogel coating could be observed clearly after soaking in PBS for 30 days no matter how the temperature changed. These results suggested that hydrogel coating can not only successfully modify PP mesh by the FT method, but also the coating exhibited a degree of stability in an aqueous environment.

3.4. Evaluation of biocompatibility in vitro

Hemocompatibility is the requirement for surface modification of modified PP mesh, as it helps to prevent hemolysis [42]. Therefore, hemolysis ratio of the composite hydrogels and modified PP-based meshes was assessed by direct contact with red blood cells in the blood. Consequently, a lower hemolysis ratio means fewer red blood cell ruptures, indicating better blood compatibility. As shown in Fig. 5a, the hemolysis ratio of V-PP mesh, PVHD₇ hydrogel, and PVHD₇-c-PP mesh were <1%, which was far below the acceptable threshold of 5% (ASTM F756–2008). Further, the hemolysis ratio of all composite hydrogels and all modified PP meshes also were <1% (Figs. S3 and S4). These results suggested that the modified PP mesh possessed excellent hemocompatibility and could be potentially direct contact with the injured tissue.

In the biomedicine realm, favorable cytocompatibility is a

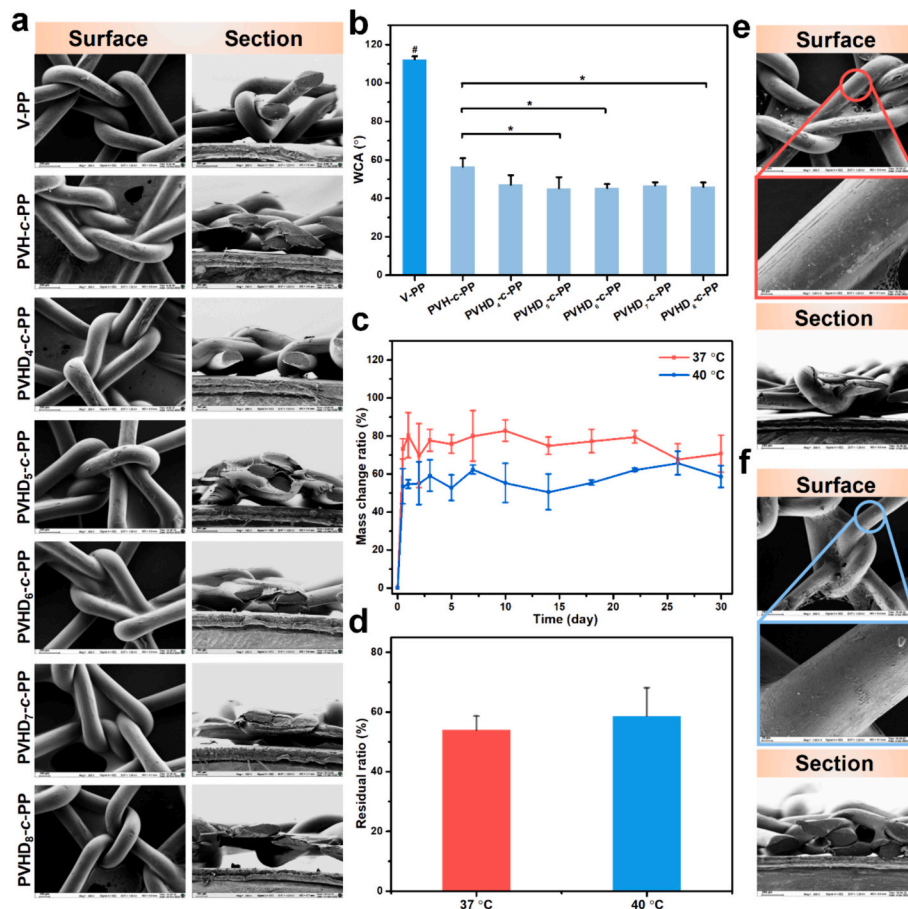


Fig. 4. (a) SEM images of V-PP and modified PP-based meshes. (b) Water contact angle (WCA) of V-PP and modified PP-based meshes. Asterisks show the results that are statically significant with P -values <0.05 (*), (#, with all other samples). (c) Mass change behavior of PVHD₇-c-PP at 37 °C and 40 °C as time prolongs. (d) The residual ratio of hydrogel coating on the PVHD₇-c-PP after immersing in PBS at 37 °C and 40 °C for 30 days. (e) SEM images of the PVHD₇-c-PP after immersing in PBS at 37 °C for 30 days. (f) SEM images of the PVHD₇-c-PP after immersing in PBS at 40 °C for 30 days.

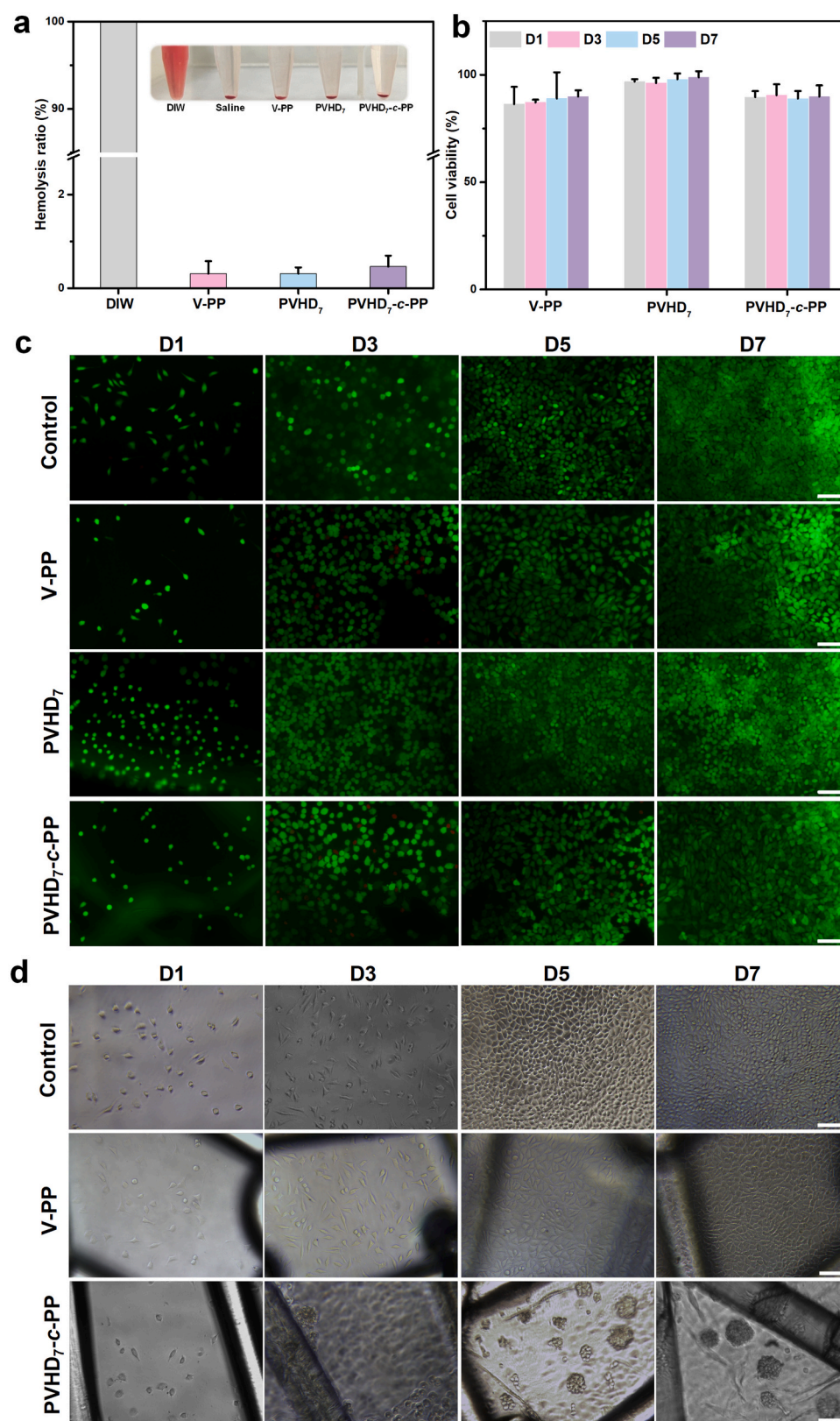


Fig. 5. (a) Hemolysis ratio of V-PP mesh, PVHD₇ hydrogel, and PVHD₇-c-PP mesh, where the deionized water (DIW) and normal saline were used as the positive and negative controls, respectively. (b) The cell viability of L929 cells after cultured with the PVHD₇ hydrogel's extraction medium, V-PP mesh and PVHD₇-c-PP mesh for 1, 3, 5, and 7 days. A normal culture medium was used as the control. (c) Live/dead cell fluorescence images of L929 cells cultured with the PVHD₇ hydrogel's extraction medium, V-PP mesh, and PVHD₇-c-PP mesh for 1, 3, 5, and 7 days. TCPS culture plates were used as the control. Scale bars represent 100 μ m. (d) Optical images of L929 cells cultured on V-PP and PVHD₇-c-PP for 1, 3, 5, and 7 days. TCPS culture plates were used as the control. Scale bars represent 100 μ m.

fundamental prerequisite for biomaterials. In order to evaluate the cytotoxicity *in vitro* of modified PP meshes, L929 cell line was selected as the tested cell line and the cell viability of L929 cells was assessed by the CCK-8 assay. After incubation for 1, 3, 5, and 7 days, the results of cell viability as illustrated in Fig. 5b. The cell viability of L929 cells did not show significant difference after culturing with these samples for 1, 3, 5, and 7 days. It is noting that the cell viability after co-culturing with PVHD₇ hydrogel's extraction medium was slightly higher than this value of co-culturing with V-PP and PVHD₇-c-PP mesh. This is due to the generally high level of safety and excellent biocompatibility of PVA and HA. Additionally, the live/dead fluorescence staining was carried out to check the status of L929 cells. As depicted in Fig. 5c, V-PP, PVHD₇ hydrogel, and PVHD₇-c-PP all presented a substantial proportion of viable cells (green fluorescence) and a negligible fraction of dead cells (red fluorescence) and the density of cells gradually increased as the incubation time prolonged. For a more comprehensive assessment of cytotoxicity, all composite hydrogels and modified PP-based meshes were tested for cytocompatibility. The results in Figs. S5, S6, S7, and S8 indicated that the proposed samples possessed outstanding cytocompatibility.

It is well established that inhibition of fibroblast adhesion to the hernia repair network is essential to prevent the formation of dense adhesions. Therefore, the L929 cells were inoculated on the surface of PP-based meshes and subsequently the morphology of cells on the material surface were observed after 1, 3, 5, and 7 days of incubation (Figs. 5d and S9). As the incubation time prolonged, the cells cultured on the surface of TPCS culture plate and V-PP mesh exhibited an adherent state. What is interesting is that the morphology of cells on the surface of PVHD₇-c-PP display obvious difference with the TPCS culture plate and V-PP mesh. In the PVHD₇-c-PP mesh group, the cells aggregated and subsequently proliferated to form multicellular spheroids. As hydrophilic polymers PVA and HA were exposed to a liquid environment, they swelled and produced a hydration layer near the surface, which prevented the proteins from adhesion. Cells cannot directly adhere to the surface, but depend on the adhesion of proteins. Thus, the presence of a hydration layer resists cell adhesion. These disclosed results

demonstrated that the modified PP mesh has excellent biocompatibility and the potential for anti-adhesion applications in hernia repair.

3.5. Evaluation of anti-adhesion property *in vivo*

3.5.1. Effect of preventing adhesion

An ideal physical barrier for anti-adhesion after surgery should be resistant to adhesion of fibroblasts. The abovementioned evaluations implied that PVHD₇-c-PP possessed excellent biocompatibility and resistance to cell adhesion. Thus, the anti-adhesive ability *in vivo* of PVHD₇-c-PP mesh was further investigated by using a rat abdominal wall defect model. At the same time, virginal PP mesh was administered as control. The abdomens of treated animals were reopened and the adhesion score was scored and measured in terms of the adhesion scoring criterion (Table S3) at specific time intervals (7, 14, and 30 days). The modeling procedure is shown in Fig. 6a. The PP-based meshes (1.5 cm × 1.5 cm) were sutured to the abdominal wall defect site for repair and anti-adhesion. After implantation of all PP-based meshes, none of the animals died and their body weight increased.

At 7, 14, 30 days after implantation, rats were sacrificed, and adhesion score and representative macroscopical observation of implantation zone were illustrated in Fig. 6b and c. On day 7 post-operation, rats developed severe adhesion between the visceral and abdominal wall was in the V-PP mesh group. On days 14 and 30 post-operation, abdominal adhesion was still detected. In contrast to the findings in V-PP mesh group, the abdominal adhesion was alleviated in PVHD₇-c-PP mesh group. In particular, no significant adhesion was noticed in the central region of PVHD₇-c-PP mesh on days 14 and 30 post-operation. Further, the degree of adhesion was quantified according to the adhesion scoring system. During the implantation period, V-PP mesh group had a significantly higher adhesion scores than that of PVHD₇-c-PP mesh group. As the presence of hydration layer on the surface of hydrogel coating, adhesion of adhesion proteins and fibroblasts was inhibited, thus preventing the formation of postoperative adhesions. Besides, the compressive modulus and softness of the PVHD₇ hydrogel coating were similar to that of natural abdominal wall, which

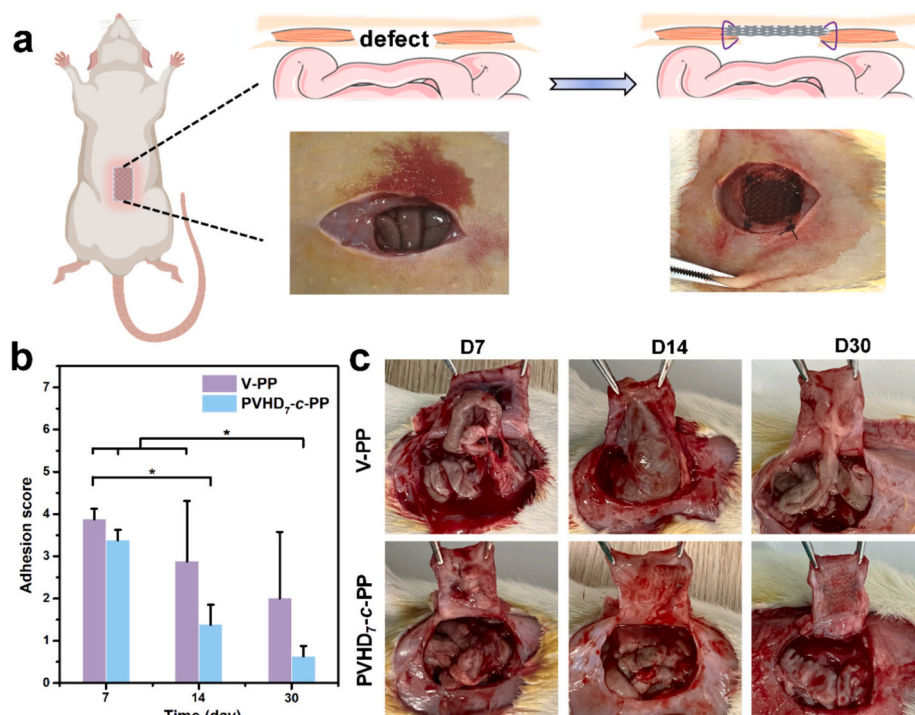


Fig. 6. (a) Schematic diagram and photos of the abdominal wall defect model and fixation of PP-based meshes in rats. (b) Adhesion score of PP-based meshes at 7, 14 and 30 days. (c) Digital photos of different PP-based mesh implanted sites after *in vivo* study on different time points.

can alleviate the stimulation of damaged tissue by the modified mesh. A comparison of the results revealed that the PVHD₇-c-PP mesh can effectively prevent bowel adhesions.

3.5.2. Histopathological and immunohistochemistry analysis

The pathological changes of surgical site were analyzed through H&E and MT staining (Fig. 7). As shown in H&E staining results, there was infiltration of numerous inflammatory cells in the tissue surrounding the PP filaments and areas of hemorrhage were also observed at 7 days of V-PP mesh implantation. On days 14 and 30 post-implantation, accumulation of inflammatory cells was still found, indicating the existence of inflammatory response in the whole observation period. However, the inflammatory cells in the surgical site of PVHD₇-c-PP mesh group were fewer and more uniformly distributed and faded away by 30 days as compared to the V-PP group, confirming the generation of a milder inflammatory reaction. Besides, new blood vessels could be observed in the PVHD₇-c-PP mesh group at day 30 implantation. The introduction of hydrogel coating, which is biocompatible and similar in softness to natural abdominal wall tissue, reduces irritation of the surgical site by the native PP, resulting in the alleviation of inflammatory response. MT staining results showed that the dense deposition of collagen fibers in the defect site of V-PP mesh group, especially on day 30 (Fig. 7b). Excessive collagen might lead to muscle atrophy and the formation of postoperative adhesion [43]. However, compared to the V-PP mesh group, significant looser collagen fibers were deposited around the PVHD₇-c-PP filaments in the whole observation period. It was worth noting that the regenerated collagen possessed regular morphology and the muscle tissue at the defect was continuous in PVHD₇-c-PP mesh group at day 30 post-implantation.

The expression levels of inflammatory cytokines (TNF- α and IL-6) cytoplasm were investigated by immunohistochemistry analysis. As shown in Fig. 8a and c, the positive immunohistochemical staining for TNF- α and IL-6 were represented by brown granules in the cytoplasm. The expression levels of TNF- α and IL-6 in V-PP mesh group were significantly higher than that in the PVHD₇-c-PP mesh group during the whole experimental durations. At the same time, semiquantitative analysis of inflammatory cytokines was performed by using Image J software. Significant differences were observed between the V-PP mesh group and PVHD₇-c-PP mesh group at each time interval postoperative. These results are consistent with the observation from adhesion evaluation, H&E staining, and MT staining. Above all, those results demonstrated that the construction of a biocompatible hydrogel coating that mimics the softness of abdominal wall tissue could not only alleviate the inflammatory response but also prevent the formation of bowel adhesions.

4. Conclusions

In this study, an anti-adhesive PP mesh with composite hydrogel

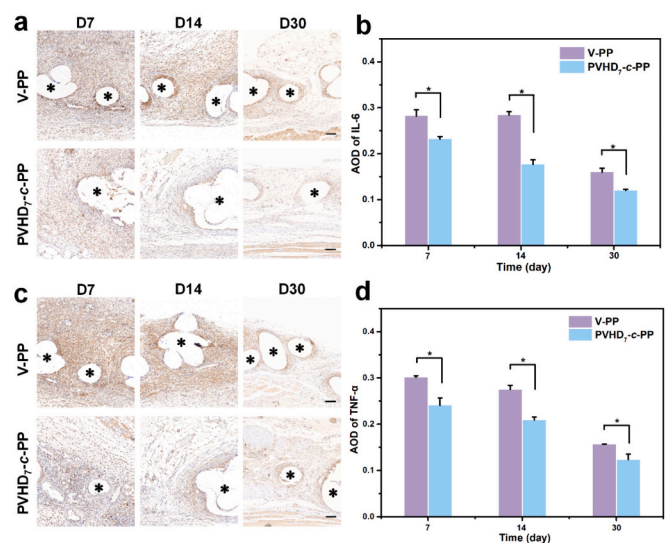


Fig. 8. (a) Micrographs of IL-6 of the treated-abdominal wall defects by immunohistochemical staining at different time points. “*” = the monofilament of the PP mesh and scale bar = 100 μ m. (b) Semiquantitative results of the IL-6 expression level. Asterisks show the results that are statically significant with P-values < 0.05 (*). (c) Micrographs of TNF- α of the treated-abdominal wall defects by immunohistochemical staining at different time points. “*” = the monofilament of the PP mesh and scale bar = 100 μ m. (d) Semiquantitative results of the TNF- α expression level. Asterisks show the results that are statically significant with P-values < 0.05 (*).

coating was successfully fabricated by simple and benign FT method. The composite hydrogel was made up of PVA and HA, and DHPA. The addition of DHPA can induce the formation of hydrogen bonds between PVA and HA molecular chains, leading to faster formation of composite hydrogels with only 3 h-long freezing at -20°C and 4 h-long thawing at room temperature. Moreover, by adding DHPA, the compressive modulus of composite hydrogel coating can also be adjusted to resemble the natural abdominal wall tissue (optimized composite hydrogel PVHD₇). *In vitro* experiments demonstrated that the PVHD₇-c-PP mesh displayed excellent biocompatibility and potential to resist L929 cells adhesion. *In vivo* anti-adhesion tests illustrated that PVHD₇-c-PP mesh could effectively alleviate inflammatory response, and reduce IL-6 and TNF- α expression. In addition, the deposition of collagen fibers was loose and regular around the PVHD₇-c-PP filaments compared to original PP. Overall, this type of modified PP mesh with composite PVHD₇ hydrogel coating, prepared by a repeatable and benign method, exhibits effective anti-adhesive property, thus being a promising prosthetic material for hernia repair.

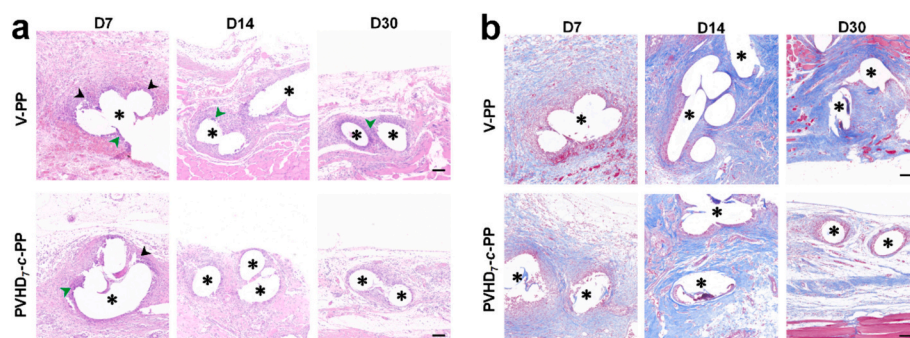


Fig. 7. Histological observation of treated-abdominal wall defects stained with (a) H&E staining and (b) MT staining at different time points. Black arrows represent neutrophils. Green arrows represent macrophages. Collagen was stained blue by MT staining. “*” = the monofilament of the PP mesh. Scale bar = 100 μ m. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

CRediT authorship contribution statement

Dandan Wei: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Yulin Huang:** Resources, Methodology. **Min Liang:** Methodology, Investigation. **Liuxin Yang:** Methodology. **Guanhua Jiao:** Investigation. **Yinghua Tao:** Methodology. **Li Xu:** Investigation. **Tianzhu Zhang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Zhenling Ji:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijbiomac.2024.132061>.

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